National University of Singapore

Bridging Mathematics

Semester 1, AY2025/2026

Time allowed: 1 hour and 30 minutes

Mock Test
Sample Solutions

Question 1 [15 marks]

(a) Write down the derivative of $y = \tan x$. Hence, or otherwise, compute the derivative of the inverse tangent function $y = \tan^{-1} x$.

(b) Use integration by parts to evaluate $\int \tan^{-1} x \, dx$.

(c) Determine the first five terms in the Maclaurin series for $y = \tan^{-1} x$.

(a) We have

$$\frac{d}{dx} \tan x = \sec^2 x.$$

Let $y = \tan^{-1} x$. Then, $\tan y = x$—we differentiate both sides of the equation:

$$\frac{d}{dx} \tan y = \frac{d}{dx} x \quad \Longrightarrow \quad \sec^2 y \cdot \frac{dy}{dx} = 1 \quad \Longrightarrow \quad \frac{dy}{dx} = \frac{1}{\sec^2 y}.$$

Since $\sec^2 y = 1 + \tan^2 y$, we have

$$\frac{dy}{dx} = \frac{1}{1 + \tan^2 y} = \frac{1}{1 + x^2}.$$

(b) Let $u = \tan^{-1} x$, and $dv = dx$. Then,

$$du = \frac{1}{1 + x^2} \, dx, \quad v = x.$$

Thus,

$$\int \tan^{-1} x \, dx = \tan^{-1} x \cdot x - \int x \cdot \frac{1}{1 + x^2} \, dx.$$

We now integrate by substitution: let $t = 1 + x^2$, $dt = 2x \, dx$. Then,

$$\int x \cdot \frac{1}{1 + x^2} \, dx = \int \frac{1}{t} \cdot \frac{1}{2} dt = \frac{1}{2} \ln t + c = \frac{1}{2} \ln \left(1 + x^2\right) + c.$$

Hence,

$$\int \tan^{-1} x \, dx = x \tan^{-1} x - \frac{1}{2} \ln \left(1 + x^2\right) + c.$$

(c) We can use the Maclaurin series expansion of $f(x) = 1/(1 - x)$:

$$\frac{1}{1 - x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots .$$

Replacing x with $-x^2$, we have

$$\begin{aligned} \frac{1}{1 - (-x^2)} &= 1 + \left(-x^2\right) + \left(-x^2\right)^2 + \left(-x^2\right)^3 + \cdots \\ &= 1 - x^2 + x^4 - x^6 + x^8 - \cdots . \end{aligned}$$

Since $f(x)$ is the derivative of $\tan^{-1} x$, we can integrate this expression to obtain the Maclaurin series of the inverse tangent:

$$\begin{aligned} \tan^{-1} x = \int \frac{1}{1 - (-x^2)} \, dx &= \int 1 - x^2 + x^4 - x^6 + x^8 - \cdots \\ &= x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \frac{x^9}{9} + \cdots + c. \end{aligned}$$

When $x = 0$, we have $\tan^{-1}(0) = 0$, and so $c = 0$. Hence,

$$\tan^{-1} x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n + 1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \frac{x^9}{9} + \cdots .$$

Question 2 [10 marks]

Consider the indefinite integral

$$\int \frac{1}{(x-a)(x-b)}\,dx,$$

where a, b are some real numbers.

(a) Evaluate the integral if $a \neq b$.

(b) Evaluate the integral if $a = b$.

(a) We perform a partial fraction decomposition:

$$\frac{A}{x-a} + \frac{B}{x-b} = \frac{A(x-b) + B(x-a)}{(x-a)(x-b)} = \frac{(A+B)x - Ab - Ba}{(x-a)(x-b)}.$$

We thus require that $A + B = 0$, and $-Ab - Ba = 1$. Since $B = -A$, we have

$$-Ab - (-A)a = A(a-b) = 1 \quad \Longrightarrow \quad A = \frac{1}{a-b},\ B = -\frac{1}{a-b}.$$

Thus, the integral becomes

$$\begin{aligned}\int \frac{1}{(x-a)(x-b)}\,dx &= \frac{1}{a-b}\int \frac{1}{x-a} - \frac{1}{x-b}\,dx \\ &= \frac{1}{a-b}\left(\ln|x-a| - \ln|x-b|\right) + c.\end{aligned}$$

(b) When $a = b$, the integral reduces to

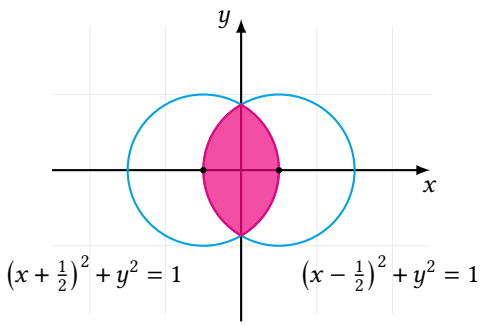
$$\int \frac{1}{(x-a)^2}\,dx.$$

We integrate by substitution: let $u = x - a$, then $du = dx$, and so

$$\int \frac{1}{(x-a)^2}\,dx = \int \frac{1}{u^2}\,du = -\frac{1}{u} + c = -\frac{1}{x-a} + c.$$

Question 3 [10 marks]

Two circles of radius 1 are made to overlap such that the center of each circle lies on the boundary of the other, as shown in the diagram below.



Determine the volume of the solid obtained by rotating the overlapping region about the x-axis.

Consider the section of the shaded region when $x > 0$: observe that

$$y = \sqrt{1-\left(x+\frac{1}{2}\right)^2} = \sqrt{-x^2-x+\frac{3}{4}}.$$

Thus, the volume obtained when this region is rotated about the x-axis is given by

$$\begin{aligned}\int_0^{1/2} \pi y^2\,dx &= \pi\int_0^{1/2} -x^2-x+\frac{3}{4}\,dx \\ &= \pi\left[-\frac{1}{3}x^3-\frac{1}{2}x^2+\frac{3}{4}x\right]_0^{1/2} \\ &= \pi\left(-\frac{1}{24}-\frac{1}{8}+\frac{3}{8}\right) = \frac{5}{24}\pi.\end{aligned}$$

Since the solid we are interested in is symmetric about the y-axis, the total volume of the solid is double the above value: $5\pi/12$.

Question 4 [15 marks]

Determine the values of x for which the following series converges:

$$\sum_{n=1}^{\infty} \frac{1}{2^n} \tan \frac{x}{2^n}.$$

For the values of x you have found, find the corresponding sum.

Hint: You may use the fact that $\tan(x/2) = \cot(x/2) - 2\cot x$.

When $x = 0$, the series converges to 0. Thus, suppose that $x \neq 0$: the nth partial sum is given by

$$\sum_{i=1}^{n} \frac{1}{2^i} \tan \frac{x}{2^i} = \frac{1}{2} \tan \frac{x}{2} + \frac{1}{4} \tan \frac{x}{4} + \frac{1}{8} \tan \frac{x}{8} + \cdots + \frac{1}{2^n} \tan \frac{x}{2^n}.$$

Proceeding from the hint, we may rewrite each term as

$$\frac{1}{2^n} \tan \frac{x}{2^n} = \frac{1}{2^n} \left(\cot \frac{x}{2^n} - 2 \cot \frac{x}{2^{n-1}} \right).$$

The series then becomes

$$\begin{aligned}
\sum_{i=1}^{n} \frac{1}{2^i} \tan \frac{x}{2^i} &= \frac{1}{2} \left(\cot \frac{x}{2} - 2\cot x \right) + \frac{1}{4} \left(\cot \frac{x}{4} - 2 \cot \frac{x}{2} \right) + \\
&\quad \frac{1}{8} \left(\cot \frac{x}{8} - 2 \cot \frac{x}{4} \right) + \cdots + \frac{1}{2^n} \left(\cot \frac{x}{2^n} - 2 \cot \frac{x}{2^{n-1}} \right) \\
&= \left(\frac{1}{2} \cot \frac{x}{2} - \cot x \right) + \left(\frac{1}{4} \cot \frac{x}{4} - \frac{1}{2} \cot \frac{x}{2} \right) + \\
&\quad \left(\frac{1}{8} \cot \frac{x}{8} - \frac{1}{4} \cot \frac{x}{4} \right) + \cdots + \left(\frac{1}{2^n} \cot \frac{x}{2^n} - \frac{1}{2^{n-1}} \cot \frac{x}{2^{n-1}} \right) \\
&= -\cot x + \frac{1}{2^n} \cot \frac{x}{2^n}.
\end{aligned}$$

We consider the limit of partial sums:

$$\begin{aligned}
\lim_{n\to\infty} \left(-\cot x + \frac{1}{2^n} \cot \frac{x}{2^n} \right) &= -\cot x + \lim_{n\to\infty} \frac{\cot(x/2^n)}{2^n} \cdot \frac{x}{x} \\
&= -\cot x + \lim_{n\to\infty} \frac{\cos(x/2^n)}{x} \cdot \frac{x/2^n}{\sin(x/2^n)} \\
&= -\cot x + \frac{1}{x} \cdot 1 = \frac{1}{x} - \cot x.
\end{aligned}$$

Hence, the series converges to the above sum for all $x \neq 0$ and $x \neq k\pi$, where k is an integer.

Question 5 [10 marks]

(a) Write down the Maclaurin series for $f(x) = \cos\left(x^3\right)$.

(b) Hence, or otherwise, determine the value of $f^{(18)}(0)$. You may leave your answer in terms of factorials.

(a) We directly use the Maclaurin series for $\cos x$, replacing x with x^3 :

$$\cos\left(x^3\right) = \sum_{n=0}^{\infty} (-1)^n \frac{\left(x^{2n}\right)^3}{(2n)!} = 1 - \frac{x^6}{2!} + \frac{x^{12}}{4!} - \frac{x^{18}}{6!} + \cdots .$$

(b) Observe that, instead of repeatedly computing the derivative of f, we may simply consider the corresponding term in its power series representation. In particular, the term in x^{18} is given by

$$\frac{f^{(18)}(0)}{18!} x^{18}.$$

From the Maclaurin series expansion in the previous part, we know that this term is simply

$$\frac{f^{(18)}(0)}{18!} x^{18} = -\frac{x^{18}}{6!}.$$

Equating coefficients, we can conclude that

$$f^{18}(0) = -\frac{18!}{6!}.$$

Question 6 [15 marks]

Let $\vec{v}_1$ be a unit vector, and $\vec{v}_2$ be a vector of length 10 such that $\vec{v}_1 \cdot \vec{v}_2 = 5$. We define a *sequence* of vectors:

$$\vec{v}_3 = \text{proj}_{\vec{v}_1}\vec{v}_2, \quad \vec{v}_4 = \text{proj}_{\vec{v}_2}\vec{v}_3, \quad \ldots, \quad \vec{v}_n = \text{proj}_{\vec{v}_{n-2}}\vec{v}_{n-1}.$$

(a) Compute $|\vec{v}_3|$, $|\vec{v}_4|$, and $|\vec{v}_5|$.

(b) Hence, or otherwise, show that the following infinite series converges, and compute its sum:

$$\sum_{n=1}^{\infty} |\vec{v}_n|.$$

(a) We first compute $\vec{v}_3$:

$$\vec{v}_3 = \text{proj}_{\vec{v}_1}\vec{v}_2 = \frac{\vec{v}_1 \cdot \vec{v}_2}{|\vec{v}_1|^2}\vec{v}_1 = \frac{5}{1}\vec{v}_1 = 5\vec{v}_1.$$

Thus, its length is given by

$$|\vec{v}_3| = 5\,|\vec{v}_1| = 5 \cdot 1 = 5.$$

We now compute $\vec{v}_4$:

$$\vec{v}_4 = \text{proj}_{\vec{v}_2}\vec{v}_3 = \frac{\vec{v}_2 \cdot \vec{v}_3}{|\vec{v}_2|^2}\vec{v}_2 = \frac{\vec{v}_2 \cdot 5\vec{v}_1}{|\vec{v}_2|^2}\vec{v}_2 = \frac{5^2}{10^2}\vec{v}_2 = \frac{1}{4}\vec{v}_2.$$

Hence, we have its length

$$|\vec{v}_4| = \frac{1}{4}\,|\vec{v}_2| = \frac{5}{2}.$$

We now compute $\vec{v}_5$:

$$\vec{v}_5 = \text{proj}_{\vec{v}_3}\vec{v}_4 = \frac{\vec{v}_3 \cdot \vec{v}_4}{|\vec{v}_3|^2}\vec{v}_3 = \frac{5\vec{v}_1 \cdot \frac{5^2}{10^2}\vec{v}_2}{5^2}\,(5\vec{v}_1) = \frac{5^3}{10^2}\vec{v}_1 = \frac{5}{4}\vec{v}_1$$

Hence, its length is

$$|\vec{v}_5| = \frac{5}{4}\,|\vec{v}_5| = \frac{5}{4}.$$

(b) We observe that, after the first two vectors, the lengths of the remaining vectors in the sequence form a pattern:

$$5, \quad 5 \cdot \frac{1}{2}, \quad 5 \cdot \left(\frac{1}{2}\right)^2, \quad \ldots.$$

This is just a geometric sequence with first term $a = 5$ and common ratio $r = 1/2$. Thus,

$$\begin{aligned}\sum_{n=1}^{\infty} |\vec{v}_n| &= 1 + 10 + \left(5 + \frac{5}{2} + \frac{5}{4} + \cdots\right) \\ &= 11 + \frac{5}{1 - (1/2)} = 21.\end{aligned}$$

Question 7 [10 marks]

A pyramid in xyz-space can be described by the equation

$$|x| + |y| + |z| \leq 1, \text{ where } z \geq 0.$$

You are in a helicopter viewing the pyramid from above, directly facing a side of the pyramid given by the plane

$$\Pi : \; x + y + z = 1.$$

(a) Suppose that your helicopter is situated at the point $P\,(3, 4, 10)$. Determine your distance from Π.

(b) Find the vector equation of the line ℓ normal to Π through the point P.

(c) Determine the coordinates of the point on Π that is closest to P.

(a) The plane Π has normal vector $\vec{n} = \vec{i} + \vec{j} + \vec{k}$ and contains the point $(1, 0, 0)$. Hence, the distance from a point to the plane is given by

$$d\,(P, \Pi) = \frac{\left|\vec{n} \cdot \vec{PQ}\right|}{|\vec{n}|} = \frac{\left|\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 3-1 \\ 4-0 \\ 10-0 \end{pmatrix}\right|}{\sqrt{1^2 + 1^2 + 1^2}} = \frac{2 + 4 + 10}{\sqrt{1^2 + 1^2 + 1^2}} = \frac{16}{\sqrt{3}}.$$

(b) This is simply

$$\vec{r}\,(t) = \begin{pmatrix} 3 \\ 4 \\ 10 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

(c) This is the intersection of the normal line found in the previous part with the plane equation: plugging in $x = 3 + t$, $y = 4 + t$, and $z = 10 + t$, we have

$$(3 + t) + (4 + t) + (10 + t) = 1 \quad \Longrightarrow \quad t = -\frac{16}{3}.$$

Thus, the point on Π closest to P is

$$\vec{r}\left(-\frac{16}{3}\right) = \begin{pmatrix} 3 \\ 4 \\ 10 \end{pmatrix} - \frac{16}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -7/3 \\ -4/3 \\ 14/3 \end{pmatrix}.$$

Question 8 [15 marks]

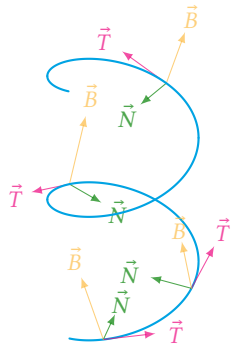
A helical curve is parametrized by the vector function

$$\vec{r}(t) = t\,\vec{i} + \cos t\,\vec{j} + \sin t\,\vec{k}, \text{ for } t \in \mathbb{R}.$$

(a) Compute the *unit tangent vector* $\vec{T}(t) = \dfrac{\vec{r}\,'(t)}{|\vec{r}\,'(t)|}$.

(b) Compute the *unit normal vector* $\vec{N}(t) = \dfrac{\vec{T}\,'(t)}{\left|\vec{T}\,'(t)\right|}$.

(c) Compute the *binormal vector* $\vec{B}(t) = \vec{T}(t) \times \vec{N}(t)$, and verify that $\vec{B}(t)$ is a unit vector.



These three vectors form an frame of reference made up of orthogonal vectors at each point on the curve $\vec{r}(t)$.

(a) The tangent vector is

$$\vec{r}\,'(t) = \begin{pmatrix} 1 \\ -\sin t \\ \cos t \end{pmatrix} \implies |\vec{r}\,'(t)| = \sqrt{1^2 + (-\sin t)^2 + (\cos t)^2} = \sqrt{2}.$$

Thus, the unit tangent vector is

$$\vec{T}(t) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -\sin t \\ \cos t \end{pmatrix}.$$

(b) The normal vector is

$$\vec{T}\,'(t) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -\cos t \\ -\sin t \end{pmatrix} \implies \left|\vec{T}\,'(t)\right| = \sqrt{0 + \left(-\frac{\cos t}{\sqrt{2}}\right)^2 + \left(-\frac{\sin t}{\sqrt{2}}\right)^2} = \frac{1}{\sqrt{2}}.$$

Thus, the unit normal vector is

$$\vec{N}(t) = \begin{pmatrix} 0 \\ -\cos t \\ -\sin t \end{pmatrix}.$$

(c) The binormal vector is

$$\vec{T}(t) \times \vec{N}(t) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -\sin t \\ \cos t \end{pmatrix} \times \begin{pmatrix} 0 \\ -\cos t \\ -\sin t \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \sin t \\ -\cos t \end{pmatrix}.$$

Observe that

$$\left|\vec{B}(t)\right| = \sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{\sin t}{\sqrt{2}}\right)^2 + \left(\frac{-\cos t}{\sqrt{2}}\right)^2} = \sqrt{\frac{1}{2} + \frac{1}{2}} = 1,$$

and indeed the vector is of unit length.

You've made it to the end!